

Compositional monitoring of energy-mix drift on the simplex

Which carrier did the structural work?

Not only which carrier got bigger — which carrier moved the composition.

Operationalizing compositional analysis — a runnable standard for researchers and the AI assistants they choose.

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Community folder: CODA-Association/ · Hand-out in UN-6 locales (EN · FR · ES · RU · ZH · AR)

The instrument reads. The expert decides. The hashes carry the receipts. The vocabulary holds the line.

The size view hides the work

A carrier can be small in share and large in structural work.

World electricity, 25 years — the standard stacked-area view

Coal stays dominant. Gas grows. Nuclear declines.
Solar appears as a thin yellow sliver after 2010.
Wind grows steadily but stays visually small until late.

What the size view misses

USA Solar, 2012 → 2013

starting share	0.107 %
structural Power Share	81.7 %
Activation Coefficient	$\approx 760 \times$

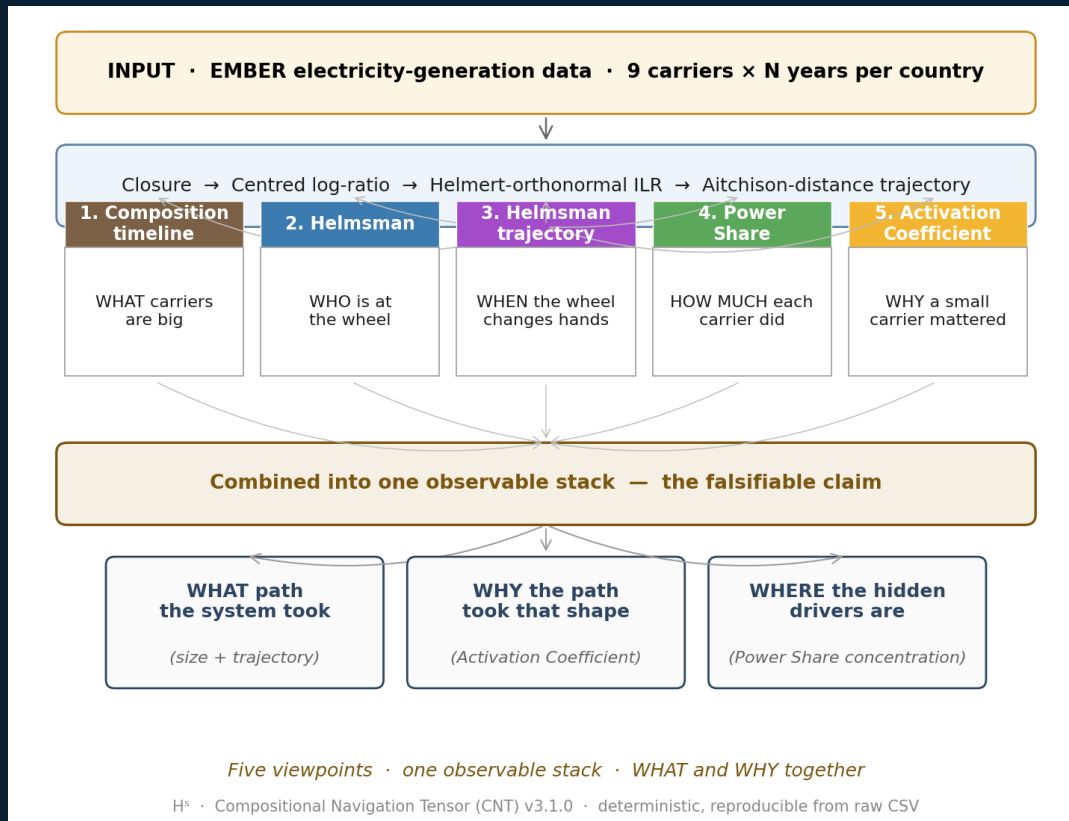
*Solar acted at $760 \times$ its size.
No size view shows that.*

This talk is the reason that number exists.

Mathematics: standard compositional data analysis. Application: monitoring frame may be new.

Five viewpoints, one observable stack

Each viewpoint answers one question. Together: an auditable transition event.



Composition

what share each carrier has.

Helmsman

which carrier has the largest CLR move at a step.

Helmsman trajectory

when the steering carrier changes.

Power Share

how much squared CLR motion each carrier did.

Activation Coefficient

Power Share ÷ starting share — the yeast factor.

All five derive from CLR + ILR-Helmert. Pure CoDa geometry; no new mathematics.

The Activation Coefficient — the yeast factor

How much structural work a carrier does, relative to how much of the mix it is.

$$\alpha_i(t) = \text{Power Share}_i(t) \div \text{starting share}_i(t)$$

$\alpha \approx 1$ carrier does work proportional to its size — ordinary

$\alpha \gg 1$ carrier acts far above its size — hidden driver

$\alpha < 1$ carrier carries less work than its size suggests — coasting

Worked example — USA Solar 2012 → 2013

starting share	0.107 %	small
Power Share	81.7 %	most of the work
α	$\approx 760 \times$	yeast

Yeast is 2% of a loaf by mass and does 100% of the rising. Same shape.

Solar 2010–2015 appears repeatedly as small-share / large-structural-work across the corpus.

Three archetypes — one instrument, three regimes

Same protocol applied to three transitions that look fundamentally different.

Germany

deliberate transition

continuous arc

Energiewende
2000 → 2025
solar + wind absorb
structural work
before size dominates.

Japan

external shock

loop and reorganise

Fukushima 2011
displaces nuclear,
causes 2011–2013
multi-year compositional
reorganisation.

United Kingdom

regime change

jump and return

Coal exit 2012–2020
from > 30 % to < 2 %.
Wind, solar, others
absorb displaced
structural work.

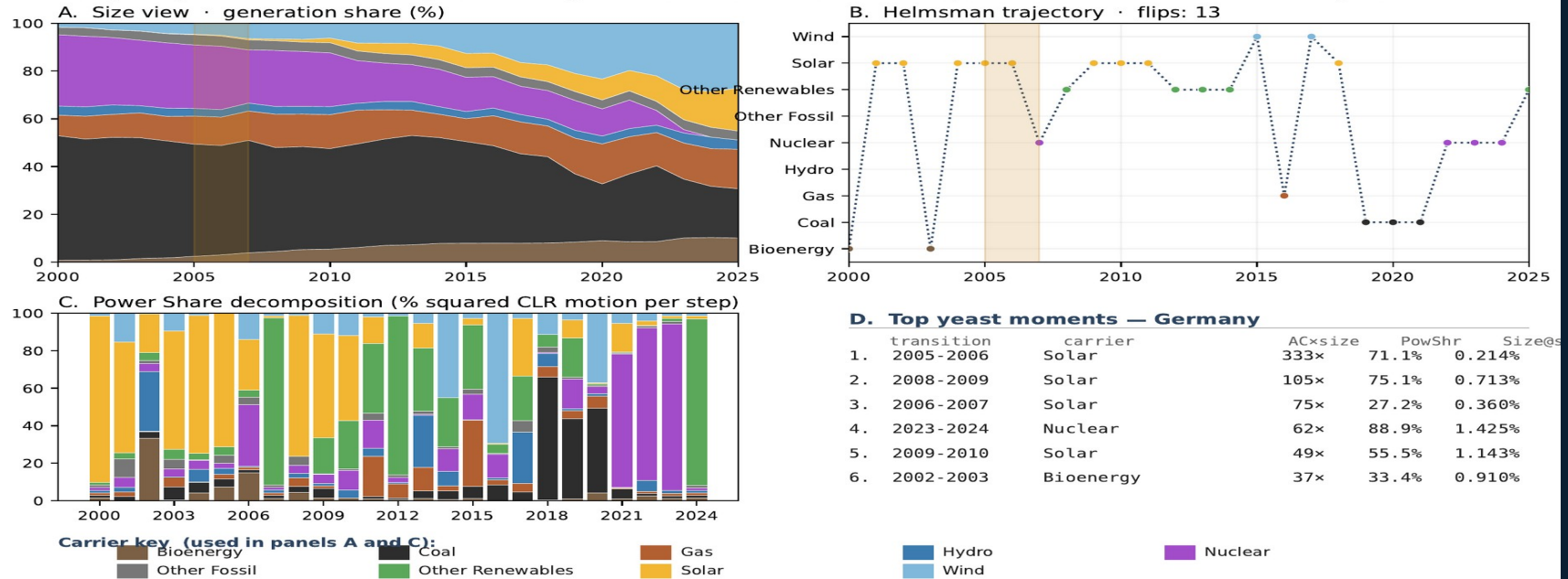
Three different transition regimes. One operational protocol reads them all.

Germany — share-and-work view

Energiewende read as a single smooth arc on the simplex (chart pair: shares + structural work).

Germany — the continuous arc to the renewable vertex

Twenty-five years of deliberate composition change. The size view shift is real — the yeast moments show when it actually accelerated.



Germany's transition is the textbook case of a continuous arc: coal recedes, gas holds, solar and wind grow steadily.

But the Power Share view shows the acceleration moments — solar in 2005-2006 at 0.21 % composition share doing 71 % of the structural directional work (activation coefficient $\approx 333\times$).

That is the structural beginning of the Energiewende — three years before the size view would call it visible.

Hidden Drivers in Energy Mixes · H² CNT v3.1.0 · Aitchison-geometric protocol

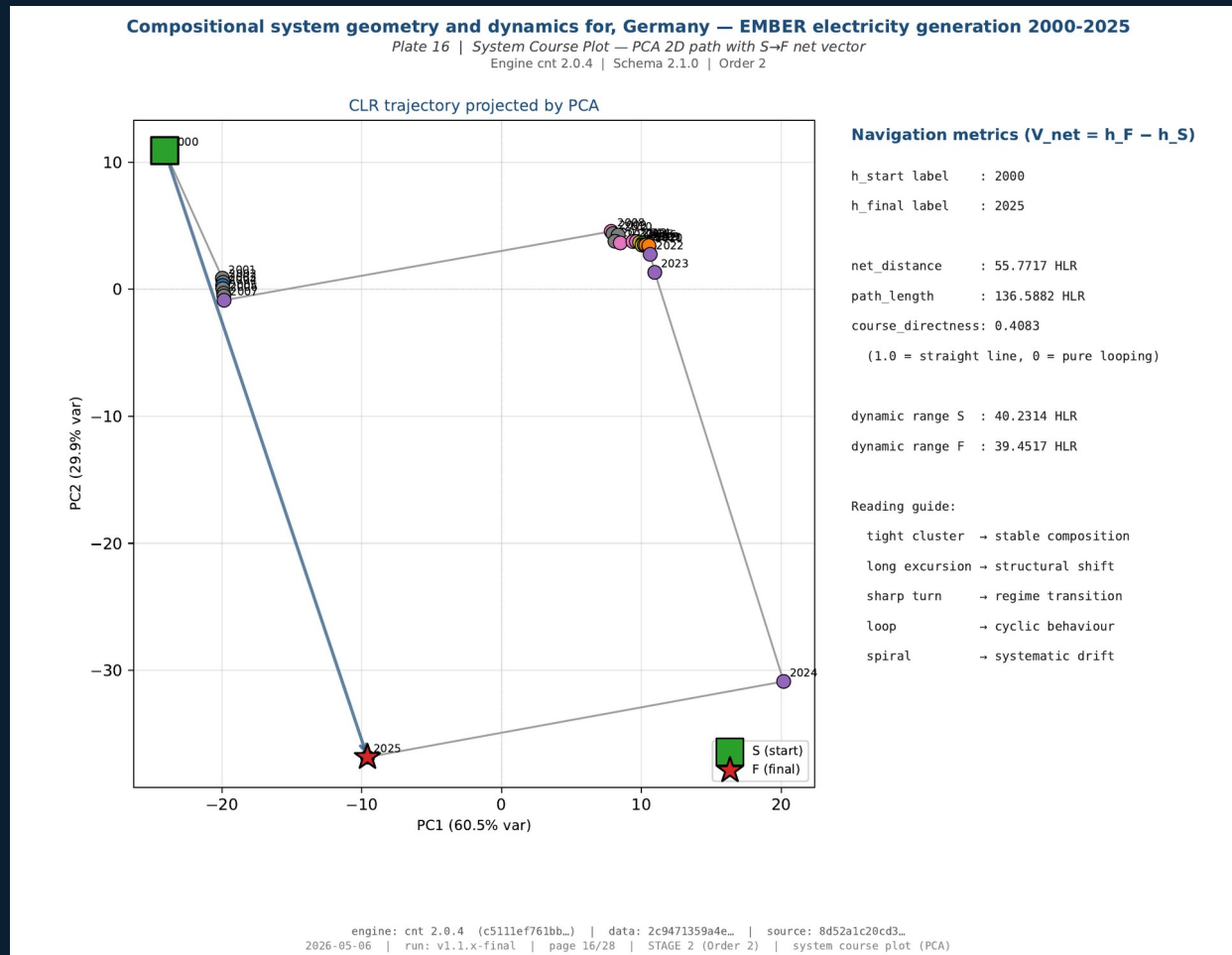
Fig. 2

Solar 2005-2006: 0.21 % share · 71.1 % structural work · $\alpha \approx 333\times$

Tiny share, dominant work. The yeast-factor signature, four years before the share view shows anything.

Germany — course on the simplex

The Helmsman trajectory: where the composition went, year by year.



Course directness 0.41 — continuous arc toward the renewable vertex.

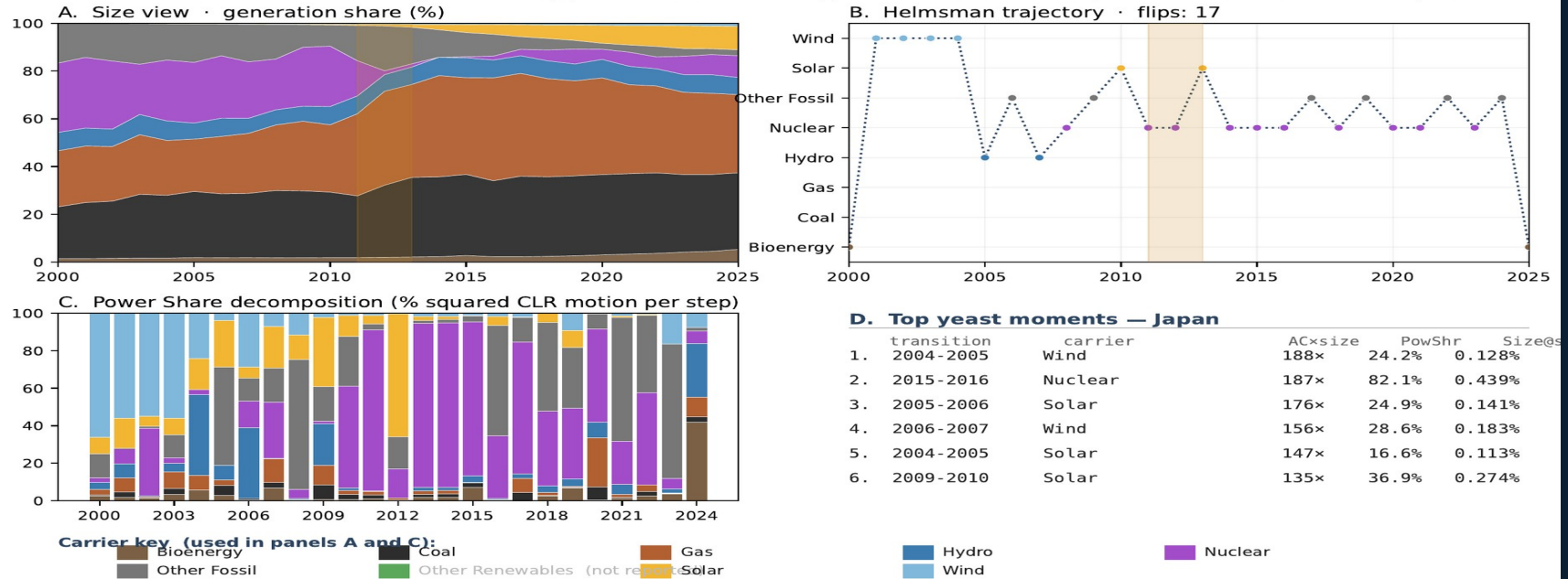
Smooth, monotone reorientation — no loops, no flips. Deliberate transition as a sustained course.

Japan — share-and-work view

Fukushima 2011: external shock displaces nuclear, multi-year compositional reorganisation.

Japan — Fukushima 2011 and the structural aftermath

An external shock and the multi-year recovery path. The helmsman flipped 17 times across the period — the loudest in the corpus.



The 2011-2012 transition shows in every viewpoint at once: composition (nuclear collapses), helmsman (carriers shift abruptly), Power Share (massive concentration of squared CLR motion at one step).

The years AFTER tell the deeper story — Japan's post-Fukushima mix kept reorganising for over a decade, and the yeast moments distribute across multiple carriers as solar, gas, and renewables each took turns absorbing the displaced nuclear share.

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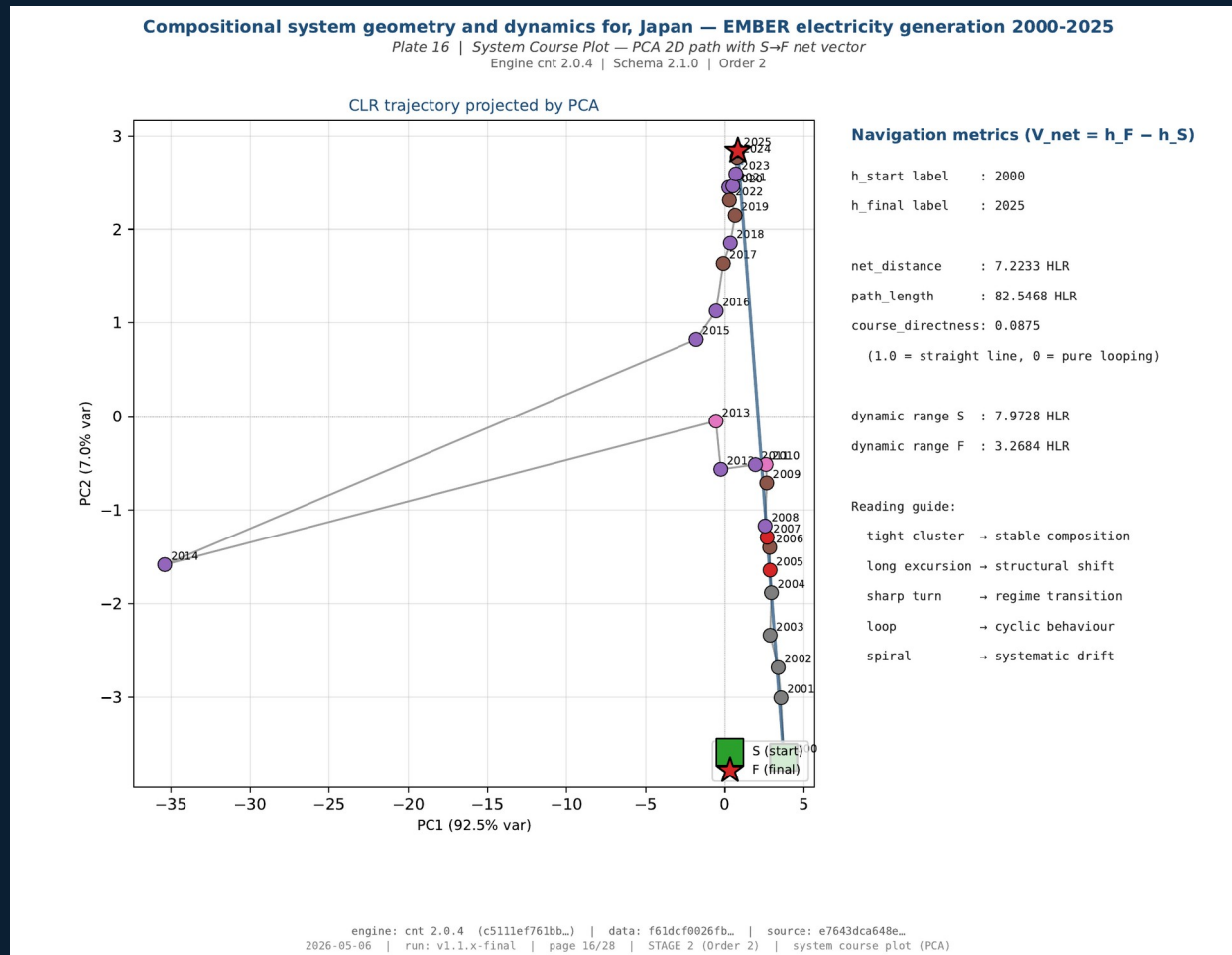
Fig. 3

Aitchison distance 2011 → 2012 ≈ 3 × neighbouring-year baseline · helmsman flips 17 ×

The instrument detects both the shock and the multi-year reorganisation that followed.

Japan — course on the simplex

The Helmsman trajectory: looping reorganisation, not a single step.



Course directness 0.09 — loop-and-reorganise archetype.

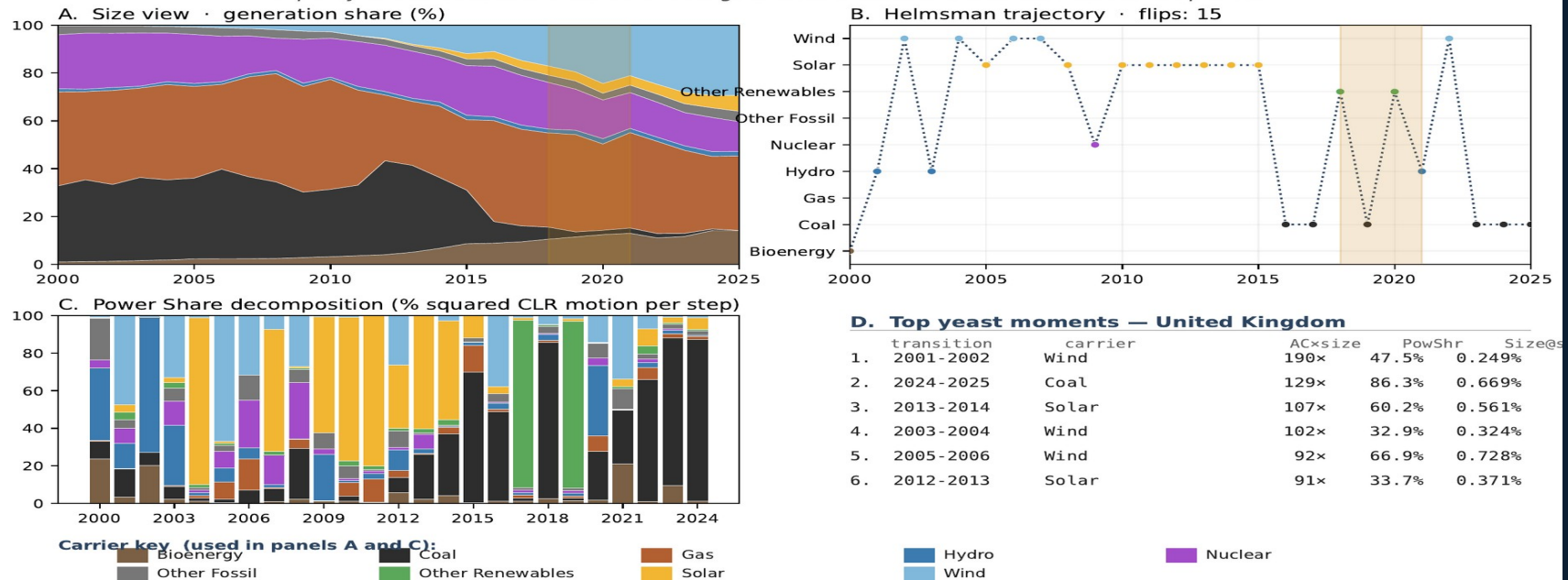
Trajectory revisits and reroutes — the system searches for a new composition, not a planned course.

United Kingdom — share-and-work view

Coal exit as policy-driven regime change, absorbed across multiple renewable carriers.

United Kingdom — the coal exit as a regime change

A policy-driven structural transition. Coal goes from dominant to near-zero across the period.



The UK story is the abrupt-policy archetype. Coal's share drops from over 30 % to under 2 % across the period.

The Power Share view names what made it possible: Other Renewables and Wind in the late-2010s took turns doing the structural work of absorbing coal's exit, often at low single-digit composition share (panel D).

Without the activation coefficient, those small-share carriers are invisible. With it, the whole transition snaps into focus.

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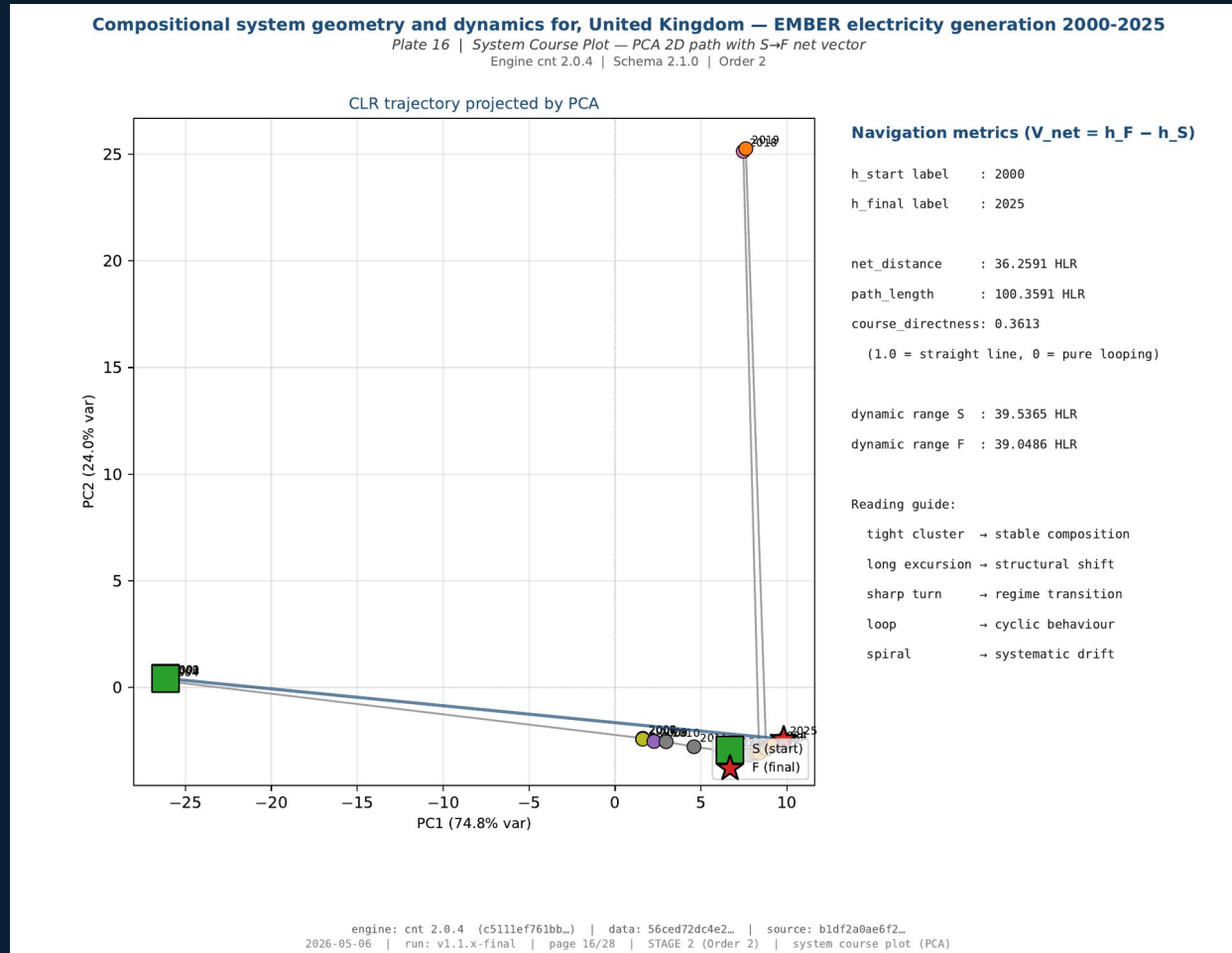
Fig. 4

Coal: > 30 % → < 2 %. Wind, solar, and other renewables absorb the displaced structural work.

Displaced work redistributes across several carriers — not one substitute, but several.

United Kingdom — course on the simplex

The Helmsman trajectory: a jump from the coal vertex, then a return toward stability.

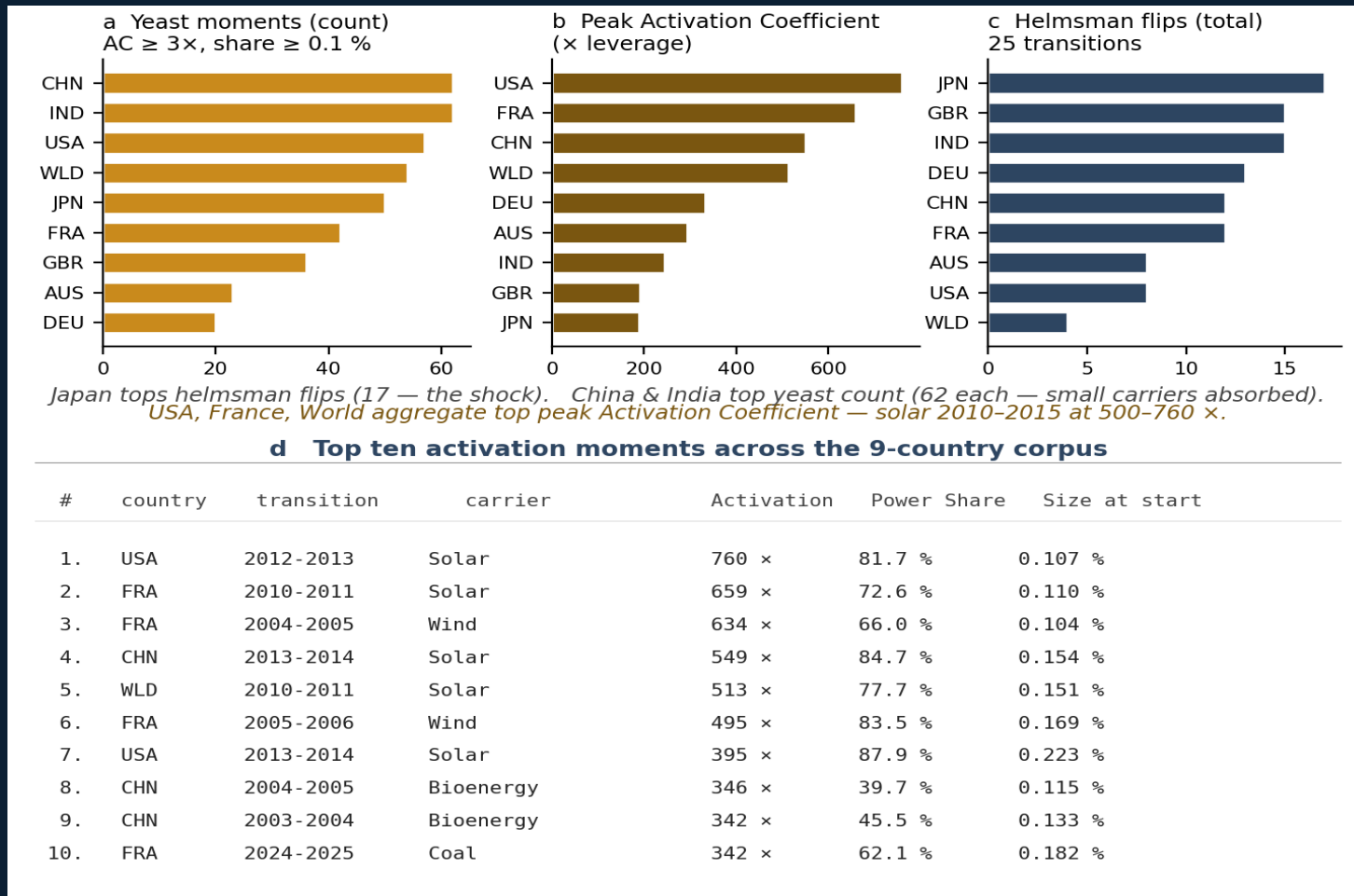


Course directness 0.36 — jump-and-return archetype.

Sharp departure from the coal vertex, then re-stabilisation — regime change as one decisive displacement.

Cross-country signature — 5 of 9 reproduce the pattern

From three case archetypes to a corpus-level result.



fires AUS • CHN • GBR • IND • JPN

quiet DEU (annual) • FRA • USA • WLD

A useful detector should not fire everywhere. Discrimination is itself evidence.

What the stack answers

One observable, five distinct questions, one reproducible object.

WHAT	carriers are big.	<i>size view</i>
WHO	is at the wheel.	<i>Helmsman</i>
WHEN	the steering changes.	<i>Helmsman trajectory</i>
HOW MUCH	work each carrier did.	<i>Power Share</i>
WHY	a small carrier mattered.	<i>Activation Coefficient</i>

The stack does not replace interpretation. It gives interpretation a reproducible object.